

The Serotonin Operating System Manual

C and x86 ASM documentation for interacting with the Serotonin kernel and userspace

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1 Preamble

This is a manual and is intended to supplement documentation. This manual will not explain every single function available in the operating system, you should also consult documentation for writing programs for Serotonin.

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2 Hardware Support

2.1 Minimum Requirements

The following specifications are required to run Serotonin:

- x86 CPU with Streaming SIMD Extensions 2 (SSE2) instruction set extension (Intel Pentium 4, AMD Athlon64, VIA C7 or newer)
- $\geq 1\text{GiB}$ of RAM
- Legacy BIOS with VESA BIOS Extensions 2.0 support
- $\geq \text{TBDGiB}$ of hard disk space

2.2 Video

Serotonin supports the VESA BIOS Extensions, preconfigured for a resolution of 1280x800 with 32-bit true colour (16,777,216 possible colours, excluding the alpha channel).

2.3 Audio

2.3.1 PC Speaker

The PC Speaker is capable of generating basic square waves.

2.3.2 Yamaha YM3812 (OPL2)

The YM3812 is a sound integrated circuit, popularised by the AdLib and Sound Blaster sound cards. It is capable of 9 melodic voice polyphony (6 melodic + 5 percussion in rhythm mode). Each voice has 2 operators for Frequency Modulation (FM) and additive synthesis, 4 selectable waveforms per operator (sine, half sine, absolute sine, quarters ine) and per operator 4-bit ADSR (attack, decay, sustain, release).

2.3.3 Sound Blaster 16

TBD

2.4 Storage

2.4.1 IDE

Serotonin currently only has support for legacy IDE ATA hard disks.

2.4.2 Filesystems

The following filesystems are supported:

- FAT32
- TMPFS/RAMFS

3 Compilation

3.1 Application Binary Interface

Serotonin uses the System V ABI, this includes the calling convention and executable format (ELF). Therefore, it is recommended to use the GNU Compiler Collection (GCC) when compiling both the kernel and usermode programs.

3.2 GCC Cross Compilation

You should use the i686-elf target for cross compiling from a different operating system other than Serotonin. This will be subject to change when an OS specific toolchain is completed at some point in the future.

4 The Serotonin Scheduler

4.1 Brief Overview

The Serotonin Scheduler is a priority round-robin scheduler for kernel mode (Ring 0, "Supervisor mode") and user mode processes (Ring 3). Kernel mode tasks are cooperative (that is, you must explicitly yield the CPU in your code), and usermode tasks are preemptive (the scheduler will

preempt your process, but will guarantee at least a quantum, currently ≥ 10 ms of CPU time). There are 256 possible priorities, 0 being the lowest (idle task) and 255 the highest (/bin/init). The scheduler will always pick the highest priority task that is ready to run. If there are multiple tasks with the same priority, it will use round-robin scheduling.

4.2 Process entry point

You must provide the entry point of the process. The scheduler will then start execution from the provided pointer, provided that the process is in the scheduler queue, the process is ready and is not blocked. You must also ensure that the loaded binary is in the correct location of memory, that is, kernel mode code should be placed in kernel mapped page area (0xC0000000 - 0xCFFFFFFF), and the equivalent for user mode (0x00400000 - 0x102FFFFFFF).

4.3 Process stack

The scheduler will allocate a stack for you, the size and location depends on the privilege level. In kernel mode processes, a stack of size 64 kibibytes is allocated, located at 0xF0400000 and growing upwards. In user mode, a stack of size 256 kibibytes is allocated, located at 0x10400000 and growing upwards.

4.4 Creating a kernel mode process

You can create a new kernel mode process by creating a process and adding it to the scheduler queue:

```
1 #include "stdio/stdio.h"
2 #include "schedule/schedule.h"
3
4 void my_process(void)
5 {
6     for (int i = 0; i < 5; i++) {
7         printf("Hello, kernel world!\n");
8         kernel_yield();
9     }
10    task_exit(EXIT_SIGTERM);
11 }
12
13 // kernel_main_high
14 process_control_block_t *my_pcb = task_create(my_process, "my_process",
15         CPU_KERNEL_MODE, 128);
16 enqueue(my_pcb);
```

Note that you have to yield the CPU after you are done with your computation. This is important so that other processes can have CPU time and are not starved. Failing to yield the CPU can result in lockups/freezes, and this is not ideal for the end user experience.

4.5 Creating a user mode process from kernel mode

While in most cases, this is not required, as you should be creating usermode tasks from usermode, it is possible to do so by loading an ELF binary.

```

1 # /bin/myusermodetask
2 .section .rodata
3 msg: .asciz "Hello, world!\n"
4
5 .section .text
6 .global _start
7 _start:
8     movl $1,    %eax # WRITE
9     movl $0,    %ebx # STDOUT
10    movl $msg, %ecx # pointer to message
11    movl $14, %edx # length
12    int $0x80      # send syscall to kernel
13
14    movl $0,    %eax # EXIT
15    movl $15, %ebx # SIGTERM
16    int $0x80      # send syscall to kernel

```

```

1 #include "stdio/stdio.h"
2 #include "schedule/schedule.h"
3 #include "kernel.h"
4
5 // kernel_main_high
6 char* elf_loc = "/bin/mytask";
7 process_control_block_t *init = task_create(NULL, elf_loc,
8     CPU_USER_MODE, 128);
9 const char *argv[1] = {"/bin/mytask"}; int argc = 1;
10 const char *envp[1] = {"PATH=/"}; int envc = 1;
11 int init_status = kernel_load_elf(init, elf_loc, elf_loc, argv, argc,
12     envp, envc);
13 kernel_yield();

```

The `kernel_load_elf` function will automatically enqueue the task into the scheduler. You can now yield from the current kernel task. Notice how there is no "yield" in the usermode task. This is because all usermode tasks are preempted by the kernel. You can read more about the Serotonin system call interface in the "System Call Interface" chapter.

4.6 Creating a user mode process from user mode

You can create a new user mode process by using the fork and execve system calls. More detail can be viewed in the "System Call Interface" chapter.

```

1  .section .text
2  .global _start
3  _start:
4      movl $4,    %eax # FORK
5      int  $0x80      # send syscall to kernel
6
7      cmpl $0,    %eax
8      je   child      # eax==0 if child
9      jg   parent      # eax==child_pid if parent (eax>0)
10
11     jmp  error        # eax<0 if fork failed
12
13 child:
14     # do whatever in child
15
16 parent:
17     # do whatever in parent
18
19 error:
20     # handle error

```

5 System Call Interface

5.1 Brief Overview

Serotonin uses the interrupt 0x80 to handle system calls generated by usermode tasks, and arguments are taken from the general purpose registers eax, ebx, ecx and edx. The eax register is used to specify the system call operation, while ebx, ecx and edx are used as arguments for the system call. For example, the exit system call uses eax=0, and ebx is used to specify the signal.

6 Kernel Compositor

6.1 Brief Overview

The Serotonin kernel provides a global compositor for all tasks, kernel and user. The compositor supports alpha blending for all z layers except for the base layer (layer zero).

6.2 Compositing in kernel tasks

6.3 Compositing in user tasks

7 Usermode C Library (Not Currently Implemented)

Not currently implemented.

8 User Interface (Not Currently Implemented)

Anything stated is planned, but subject to change.

8.1 Taskbar

Serotonin's UI contains a simple taskbar with a list of active programs, system information like CPU, GPU, RAM and Storage usage. The taskbar location can be changed in the settings, alongside its width and the colour.

8.2 Window Layout

8.2.1 Tiling Window Manager - Ionic

Serotonin implements a tiling window manager (WM) called Ionic, named after Ionic bonds. The WM is a simplistic, low fidelity UI made purely for performance. This is advised for lower end hardware nearer to the Hardware Requirements spec. Windows are organised in a grid-like structure. This WM is enabled by default in the settings.

Ionic features 2-dimensional workspaces, meaning users can have multiple virtual desktops arranged both horizontally and vertically, categorized with numbers and letters.

8.2.2 Ionic Shortcuts

Action	Shortcut
Move window	MOD + ARROW_KEY
Swap workspace (horizontal)	MOD + CTRL + ARROW_KEY
Move window between workspaces (horizontal)	MOD + SHIFT + ARROW_KEY
Swap workspace (vertical)	MOD + CTRL + PAGE UP/PAGE DOWN
Move window between workspaces (vertical)	MOD + SHIFT + PAGE UP/PAGE DOWN
Move window to workspace 1-9	MOD + SHIFT + NUM
Move window to workspace Q-E	MOD + SHIFT + LETTER

Table 1: Ionic Window Manager Shortcuts

8.2.3 Floating / Stacking Window Manager - Covalence

Covalence is a floating WM implemented into Serotonin as an alternative to Ionic. Covalence is much more intuitive for end users familiar with operating systems like MacOS, Windows and other WMs like Gnome or KDE. Named after covalence bonds, this window manager allows users to move, resize, minimize and close windows with their mouse, or optionally with keyboard shortcuts. Naturally this WM is more computationally expensive and must be enabled manually by the user.

8.3 Running Programs

8.3.1 Quick Run

Processes can be opened using the default shortcut `MOD + D` which opens the program selection dialog, [insert name here]. The user can then open a previous task using `CTRL + NUM`, where `NUM` is a number key from 1 to 3.

8.3.2 Searching Programs

The user can also begin to type the name of a program to search the system for it instead. The program is then run by pressing `ENTER`. The arrow keys or tab can be used to cycle through the current options, as well as double clicking the program in the list.

8.4 Settings

Serotonin allows users to customize their desktop experience inside the settings program. Users can change their wallpaper, taskbar colour and location, default window manager and other settings as they get implemented. The menu will render using an Ionic-style window so low end hardware can always run the settings app. This guarantees they can configure their system to perform well.